

Udowodnij, że równanie $(8 - \sqrt{15})^{\frac{1}{2}} + (8 + \sqrt{15})^{\frac{1}{2}} = \sqrt{30}$ jest prawdziwe.

$$\underbrace{(8 - \sqrt{15})^{\frac{1}{2}}}_a + \underbrace{(8 + \sqrt{15})^{\frac{1}{2}}}_b = \sqrt{30}$$

$$a^{\frac{1}{2}} = \sqrt{a}$$

① podnosimy całość do kwadratu

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(8 - \sqrt{15})^{\frac{1}{2}} + (8 + \sqrt{15})^{\frac{1}{2}} = \sqrt{30} \quad / ()^2$$

$$8 - \sqrt{15} + 2\sqrt{(8 - \sqrt{15})(8 + \sqrt{15})} + 8 + \sqrt{15} = 30$$

$$16 + 2\sqrt{64 - 15} = 30$$

$$16 + 2\sqrt{49} - 30 = 0$$

$$16 + 14 - 30 = 0$$

$$30 - 30 = 0$$

$$0 = 0$$

c.k.d.